Synthetic local-first presentation pipeline

A hypothetical design for evidence-preserving technical talks

SYNTHETIC / HYPOTHETICAL TEST MATERIAL · 9-slide backend bake-off

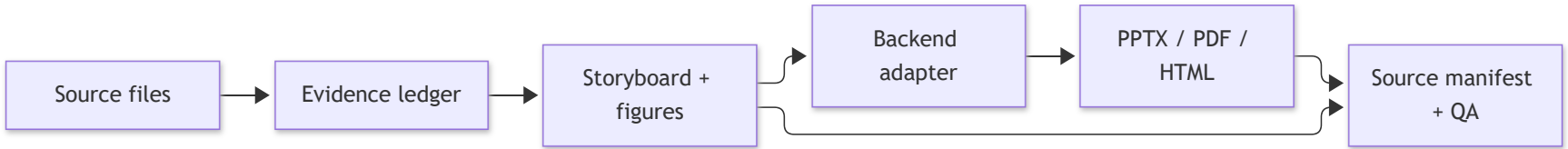
Design the pipeline around evidence

Scientific review needs exact data mapping, source lineage, and explicit uncertainty.

Presentation tools can draft and arrange slides while review checks the source trail.

> No empirical performance data

Separate evidence from presentation layout



Source identifiers and verification state travel with the rendered deck.

Keep figures deterministic

Figure field	Hypothetical input
Load tier A	120 req/s
Load tier B	240 req/s
Load tier C	360 req/s

Synthetic fixture values, not benchmark results.

Layout intent should survive backend changes

Assertion
One claim per slide

Evidence
Show source and data

Review
Keep caveats visible

Synthetic design example · no measured results

Track limitations as structured output

- Unsupported chart type → explicit warning
- Raster-only element → object-level editability report
- Missing source ID → build error
- Unverified evidence → visible pending state
- Provider call → explicit opt-in and cost boundary

Compare capabilities, not marketing labels

Check	Required evidence
Export	PPTX produced by local command
Editability	Text, shapes, charts inspected in package
Fidelity	Every rendered page reviewed
Lineage	Source IDs preserved in sidecar
Operations	Version, dependencies, services logged

A backend is useful only when its boundary is clear

- Local renderer: repeatable exports without model calls
- AI service: optional drafting with provider and cost disclosed
- Scientific QA: checks every output outside the renderer

SYNTHETIC / HYPOTHETICAL ARCHITECTURE

Recommendation for this test

Keep evidence and chart data in a backend-neutral contract.

Use isolated adapters; promote a renderer only after same-input, object-editability, and visual checks pass.

No backend comparison in this fixture establishes research performance.